

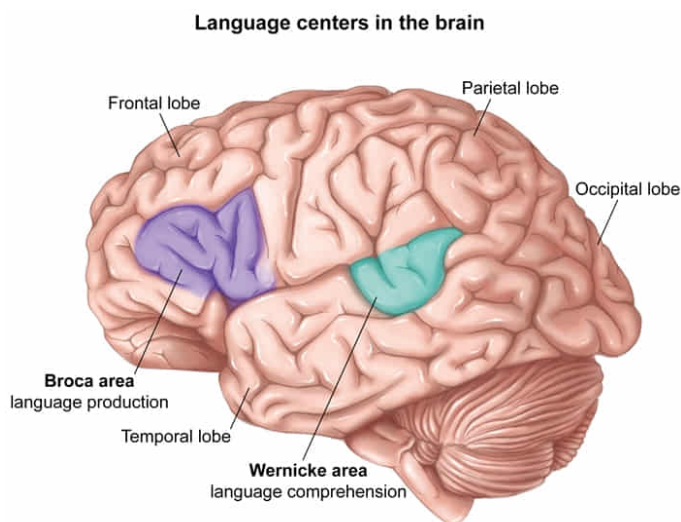
If neuroimaging studies show that auditory hallucinations are associated with increased activity in the speech production area of the brain, then PET scans of patients experiencing auditory hallucinations most likely demonstrate:

- ☐ A. increased uptake of radioactive tracer–labeled glucose in Wernicke area. [13%]
- ✓ ☒ B. increased uptake of radioactive tracer–labeled glucose in Broca area. [77%]
- ☐ C. increased deoxygenation of blood in Wernicke area. [2%]
- ☐ D. increased deoxygenation of blood in Broca area. [6%]

Correct

77% answered correctly

### Explanation:



**Language** functions in the brain tend to be **lateralized** to the **left hemisphere**. Two important language centers located in the left hemisphere are Broca and Wernicke areas.

**Broca area** (prefrontal cortex) is associated with the **production** of **expressive language** (both spoken and written). Lesions in this area (eg, due to stroke) result in difficulty producing fluent speech (Broca aphasia), although the ability to *understand* spoken and written language is largely retained.

Positron emission tomography (**PET**) is a functional imaging technique that

measures the emission of positrons that result from the metabolism of a radioactive tracer (injected prior to the scan). To measure brain activity, a radiolabeled form of glucose is used as the tracer, because glucose is metabolized more quickly by active brain regions, which are represented as hypermetabolic regions by the PET scanner.

If neuroimaging studies show that auditory hallucinations are associated with increased activity in the speech production area, then PET scans should demonstrate increased uptake of radioactive tracer–labeled glucose in Broca area.

**(Choice A)** Wernicke area (posterior temporal cortex) is associated with *language comprehension*, not speech production.

**(Choices C and D)** Functional magnetic resonance imaging (fMRI) analyzes changes in blood flow by detecting differential properties of oxyhemoglobin and deoxyhemoglobin; active brain regions require more oxygen, which are represented as regions of increased blood flow by the fMRI scanner. PET does not measure blood deoxygenation.

**Educational objective:**

Language functions in the brain tend to be lateralized to the left hemisphere. Broca area is associated with spoken and written language production, and Wernicke area is associated with language comprehension.

**Time spent:**0 sec  
**Subject:**  
Behavioral Sciences

**QID:**402359  
**System:**  
Learning, Memory, and Cognition